Mohammad Mahdi Sayyadnejad

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RESEARCH INTERESTS

Empirical Software Engineering, Mining Software Repositories (MSR), Applied NLP in Software Engineering.

EDUCATION

Sept 2021 – Sept 2024 — Shiraz, Iran

Shiraz University
MSc in Software Engineering

- *GPA:* 15.40/20
- *Thesis:* Identifying the Challenges of Developing Explainable AI Systems Using User Questions in Q&A Forums through Topic Modeling Algorithms.
- *Courses:* Advanced Database, Parallel algorithms, Model Checking, Software Architecture, Perceptual Computing, Data Mining, Information Networks.

Sept 2016 – May 2021 — Karaj, Iran

Islamic Azad University, Karaj Branch
BSc in Software Engineering

- *GPA:* 14.15/20
- *BSc Project:* Designed and developed a full-stack, decoupled mobile e-commerce platform utilizing React Native, Node.js, and Express.
- Earned A and A+ grades in key courses, including Data Mining, Operating Systems, Advanced Programming, Discrete Mathematics, and Internet Engineering.

PUBLICATIONS

Oct 2025

Exploring the Black Box: Analyzing Explainable AI Challenges and Best Practices Through Stack Exchange Discussions
Sayyadnejad, Mohammad Mahdi, Ali Asgari, Ashkan Sami, and Hooman Tahayori.
Published in Empirical Software Engineering (EMSE).
Presented at ICSE 2026 journal-first track.

- **Contribution:** First-author empirical study involving end-to-end research design, large-scale Stack Exchange data collection, NLP pipeline development, topic modeling analysis, visualization, and manuscript preparation.

Feb 2024

Identifying the Challenges of Developing an Explainable Artificial Intelligence System Using User Questions in Question-and-Answer Forums Through Text Clustering Algorithms
Sayyadnejad, Mohammad Mahdi, Ali Asgari, Ashkan Sami, and Hooman Tahayori.
20th CSI International Symposium on Artificial Intelligence and Signal Processing (AISP), 2024.

RESEARCH EXPERIENCE

June 2023 – Oct 2025 — Edinburgh, UK (Remote)

Research Assistant
Edinburgh Napier University

- Collaborated remotely with Prof. Ashkan Sami on empirical software engineering and data mining research projects.

- Engineered data extraction workflows to analyze developer data from Q&A platforms.
- Applied optimized Latent Dirichlet Allocation (LDA) models and NLP pipelines to translate unstructured text corpora into interpretable empirical findings.

June 2022 – Sept 2024 Shiraz, Iran

Research Assistant
Software Engineering Lab, Shiraz University

- Conducted large-scale empirical research under the joint supervision of Dr. Hooman Tahayori and Prof. Ashkan Sami.
- Collaborated with researchers at Delft University of Technology (TU Delft) to develop NLP and topic modeling workflows for the empirical analysis of developer discussions.
- Designed and conducted an empirical evaluation of LDA, NMF, and BERTopic for mining and analyzing 68,000+ developer discussions from Stack Exchange and GitHub, assessing topic coherence, diversity, runtime efficiency, and scalability across multiple topic configurations.
- Engineered advanced data mining and complex network analysis pipelines, implementing K-means clustering, Apriori rule generation, and Louvain community detection algorithms.
- Developed GPU-accelerated parallel processing architectures utilizing CUDA and C++ for high-performance transformations of large software engineering datasets.

PROFESSIONAL EXPERIENCE

Feb 2025 – Present Tehran, Iran

IT Engineer & Technical Research Support
Kharazmi University Faculty of Financial Sciences

- Provide technical consultation to graduate students and faculty on the AI and machine learning components of research projects, specializing in data preprocessing and model selection.
- Support research-oriented computing workflows through Python-based tooling and software environment configuration.

Nov 2022 – Mar 2023 Shiraz, Iran

Software Engineer
Abriday

- Engineered backend web services utilizing Java Spring and optimized PostgreSQL databases to support scalable data workflows in a production environment.

ACADEMIC & OPEN-SOURCE PROJECTS

LDA Topic Modeling Pipeline [Repository]
Python, NLP, Topic Modeling

- Developed an NLP pipeline for preprocessing, clustering, and analyzing large-scale textual datasets using LDA-based topic modeling.

High-Performance GPU Image Processing [Repository]
CUDA C++, Parallel Computing

- Implemented GPU-accelerated image filtering and matrix transformation algorithms using CUDA.

Intelligent Document Retrieval (RAG) [Repository]
Python, RAG Architecture, LLMs, NLP

- Engineered a Retrieval-Augmented Generation (RAG) pipeline to systematically parse, index, and query unstructured text data.

FastAPI Asynchronous Architecture [Repository]
Python, FastAPI, AsyncIO, API Design

- Developed a highly concurrent, production-ready REST API boilerplate optimized for scalable data processing and modular microservices.

SKILLS

- **Research Areas:** Empirical Software Engineering, Mining Software Repositories, SE4AI, AI4SE.
- **Research Methods:** Topic Modeling, NLP-based Empirical Analysis, Data Mining, Clustering, Association Rule Mining, Network Analysis, Explainable AI.
- **Programming:** Python, Java, SQL, JavaScript.
- **Frameworks & Libraries:** FastAPI, Spring Boot, HuggingFace, Gensim, BERTopic, spaCy, NLTK, LangGraph, LangChain, CrewAI, SHAP.
- **Data & Infrastructure:** PostgreSQL, MySQL, Data Pipelines, REST APIs.

CONFERENCES AND WORKSHOPS

Apr 2026

48th IEEE/ACM International Conference on Software Engineering (ICSE 2026)

Nov 2025

Winter School on Software Engineering & AI (ASEAI)
Workshops on artificial intelligence and empirical software engineering. [Certificate]

Aug 2025

Summer School on Data Science and AI Agents
Hertie School. Practical experience with data-driven approaches and AI agents. [Certificate]

Feb 2024

20th CSI Int'l Symposium on Artificial Intelligence and Signal Processing
Mazandaran University of Science and Technology.

June 2022

Federated Africa and Middle East Conference on Software Engineering (FAMECSE)

Feb 2022

Winter School On Software Engineering
HSE University. Empirical study, SE for researchers, and Testing. [Certificate]

CERTIFICATIONS

- **Understanding Research Methods**, *University of London* [Certificate]
- **Applied Text Mining in Python**, *University of Michigan* [Certificate]
- **Applied Machine Learning in Python**, *University of Michigan* [Certificate]
- **Generative AI Engineering and Fine-Tuning Transformers**, *IBM* [Certificate]
- **Generative AI and LLMs: Architecture and Data Preparation**, *IBM* [Certificate]
- **Fundamentals of AI Agents Using RAG and LangChain**, *IBM* [Certificate]
- **Multi-Agent Systems with LangGraph**, *Datacamp* [Certificate]
- **MLOps Fundamentals**, *Datacamp* [Certificate]
- **Google Business Intelligence**, *Google* [Certificate]